

# UNIT I

## DATA COMMUNICATION

### Characteristics of Data Communication

Four fundamental characteristics of data communication are: delivery, accuracy, timeliness, and jitter

Delivery: Delivery refers to the successful transfer of data from a source to a destination without any loss or corruption. It ensures that all the transmitted information reaches the intended recipient accurately and completely. Delivery can be affected by various factors, including network congestion, hardware failures, or software issues.

Accuracy: Accuracy refers to the correctness and integrity of the data being transmitted or received. It involves ensuring that the information remains unaltered throughout the transmission process. Data accuracy can be compromised due to factors such as noise, interference, or errors in transmission.

Timeliness: The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

Jitter: Jitter refers to the variation in the timing of data packets arriving at the destination. is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 30 ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result. Excessive jitter can result in inconsistent data delivery and can affect the quality of real-time applications, such as VoIP or video streaming.

### Components of Data Communication System

The components of a data communication system include:

Message: This is the information that needs to be communicated from one point to another.

Sender: The sender is the device or person that initiates the communication by creating and sending the message.

Receiver: The receiver is the device or person that receives the message sent by the sender.

Transmission medium: This is the physical pathway through which the message travels from the sender to the receiver. It can be wired (such as copper or fiber optic cables) or wireless (such as radio waves or infrared signals).

Protocol: A protocol is a set of rules and conventions that govern the exchange of data between the sender and receiver. It ensures that the data is transmitted and received correctly.

## Data Communication Model

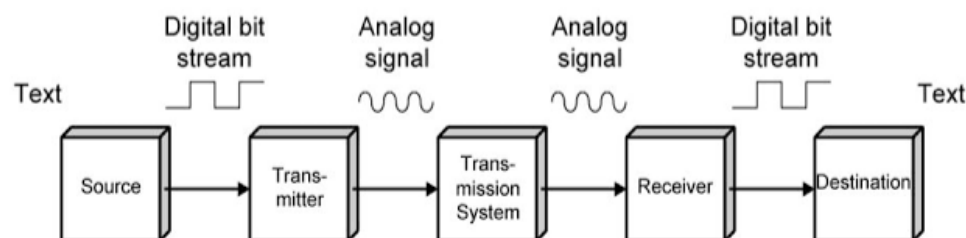
Source: This device generates the data to be transmitted; examples are telephones and personal computers.

Transmitter: the data generated by a source system are not transmitted directly in the form in which they were generated. A transmitter transforms and encodes the information in the form which is suitable for the transmission and then transmit it across a transmission system.

Transmission System: This can be a single transmission line or media that connects source and destination and share data between both entities.

Receiver: The receiver accepts the signal from the transmission system and converts it into a form that can be handled by the destination device (original form in which the data was generated by source)

Destination: Takes the incoming data from the receiver.



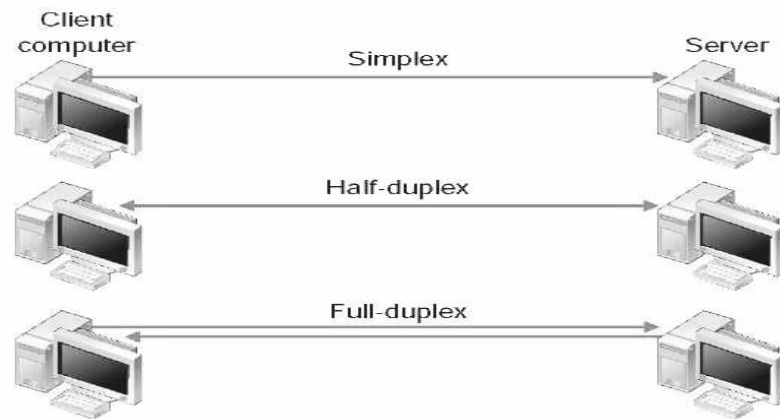
## Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex.

**Simplex**: In simplex mode, the communication is unidirectional, similar to a one-way street. Data can only be transmitted in one direction, from the sender to the receiver. Examples of simplex include loudspeakers, fire alarm systems, and television broadcasting.

**Half-Duplex**: In half-duplex mode, data can be transmitted in both directions, but not simultaneously. The communication alternates between sending and receiving data. Walkie-talkies and CB radios are examples of devices that use half-duplex mode.

**Full-Duplex:** In full-duplex mode, data can be transmitted simultaneously in both directions. This allows for two-way communication without the need to alternate between sending and receiving. Examples of full-duplex mode include telephone conversations and video conferencing.



## Data Representation

Data representation involves encoding information in a format that can be efficiently processed by computers.

Text: In data communications, text is represented as a bit pattern, a sequence of bits (0s or 1s). Different sets of bit patterns have been designed to represent text symbols. Each set is called a code, and the process of representing symbols is called coding. The prevalent coding system is called Unicode, which uses 32 bits to represent a symbol or character used in any language in the world. The American Standard Code for Information Interchange (ASCII), constitutes the first 127 characters in Unicode and is also referred to as Basic Latin.

Numbers: Numbers are also represented by bit patterns. The number is directly converted to a binary number to simplify mathematical operations.

Image: Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. Each pixel's color may be represented using RGB values or other color models like YCM/YCrCb.

Audio: Audio refers to the recording or broadcasting of sound or music. It is continuous, not discrete. Sound is represented digitally through a sequence of samples. The amplitude of each sample represents the sound wave at a specific point in time.

Video: Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.

## Analog Data and Digital Data

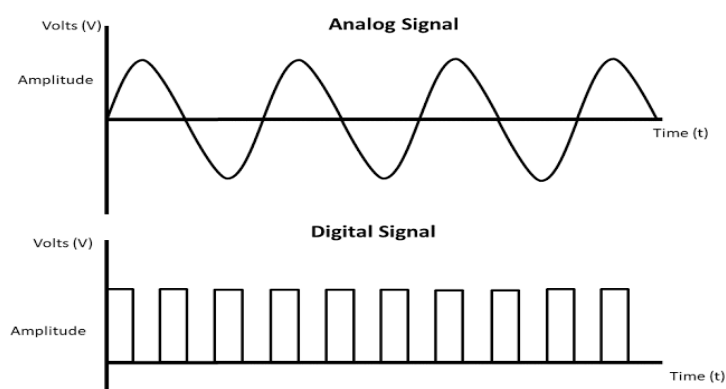
**Analog Data:** The term analog data refers to information that is continuous; For example, an analog clock that has hour, minute, and second hands gives information in a continuous form. The movements of the hands are continuous. Analog data, such as the sounds made by a human voice, take on continuous values. When someone speaks, an analog wave is created in the air. This can be captured by a microphone and converted to an analog signal or sampled and converted to a digital signal.

**Digital Data:** Digital data refers to information that has discrete states. For example, a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06. Digital data take on discrete values. For example, data are stored in computer memory in the form of 0s and 1s. It can be precisely quantized and processed by computers and digital systems. It enables precise storage, transmission, and manipulation of information.

## Signals

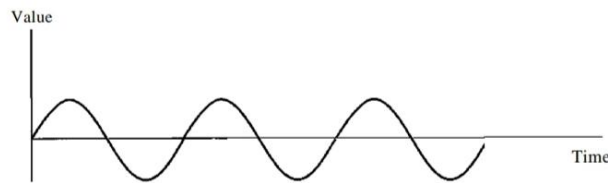
**Analog Signal:** An analog signal is any continuous-time signal representing some other quantity, i.e., analogous to another quantity. For example, in an analog audio signal, the instantaneous signal voltage varies continuously with the pressure of the sound waves. It is a form of electrical energy (voltage, current, or electromagnetic power) for which there is a linear relationship between electrical quantity and the value that the signal represents.

**Digital Signal:** A digital signal represents the original time-varying quantity as a sampled sequence of quantized values. The signal, whose amplitude takes only limited values is called a Digital signal. Digital signals are discrete, they contain only distinct values. Digital signals carry binary data i.e. 0 or 1 in the form of bits, it can only contain one value at a period of time. Digital signals are represented as square waves or clock signals.



**> Periodic and Nonperiodic Signals:** Both analog and digital signals can take one of two forms: periodic or nonperiodic. A periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods. The completion of one full pattern is called a cycle. A nonperiodic signal changes without exhibiting a pattern or cycle that repeats over time.

> **Simple Periodic Signal:** A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals.



> **Composite Signal:** A composite signal refers to a signal that combines multiple individual signals into a single signal. According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.

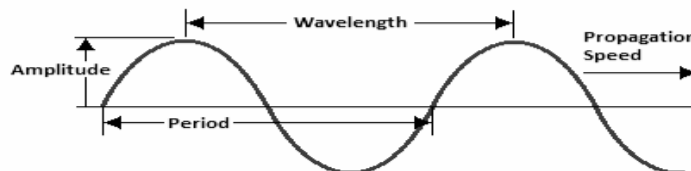
### > Parameters of Sine Wave

Frequency: The frequency of a signal is the number of cycles or repetitions of the signal that occur per unit of time. It is measured in Hertz (Hz), where 1 Hz equals one cycle per second.

Period: The period of a signal is the duration of a single cycle. It is the reciprocal of the frequency, so  $\text{period} = 1 / \text{frequency}$ . The unit of period is typically seconds.

Wavelength: Wavelength is the distance between two consecutive points of a wave that are in phase, such as two peaks or two troughs..

Amplitude: The amplitude of a signal refers to the maximum magnitude of the signal. It represents the strength or intensity of the signal.



Phase: Phase refers to the position of a point in time on a waveform cycle, relative to a reference point. It indicates how far along the waveform cycle the signal is at a given time.

## Bit Rate and Bandwidth

**Bit rate:** The number of bits per second that can be transmitted along a digital network is called bit rate or data rate. Bit rate, as the name implies, describes the rate at which bits are transferred from one location to another. It measures how much data is transmitted in a given amount of time. Bitrate is commonly measured in bits per second (bps), kilobits per second (Kbps), or megabits per second (Mbps).

**Bandwidth in Hertz:** The range of frequencies contained in a composite signal is its bandwidth. The bandwidth is normally a difference between two numbers. For example, if a

composite signal contains frequencies between 1000 and 5000, its bandwidth is 5000 - 1000, or 4000 Hz.

**Bandwidth in Bits per second:** The term bandwidth can also refer to the number of bits per second that a channel, a link, or even a network can transmit. We use the term bandwidth to describe the maximum bit rate of a system. If a network connection has a bandwidth of 100 Mbps, that means it can't transfer more than 100 megabits per second.

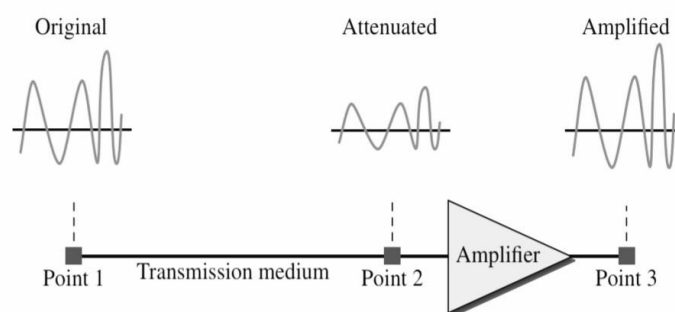
## Transmission Impairments

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise:

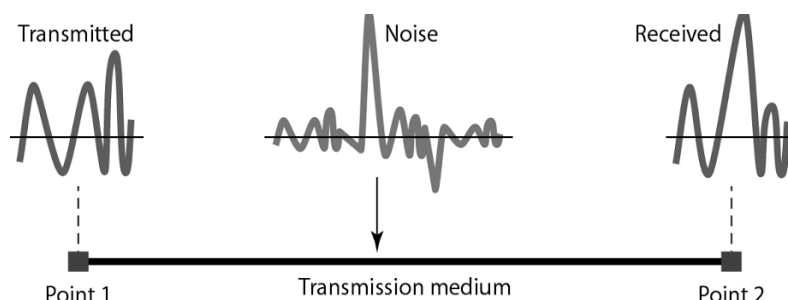
**Attenuation:** This refers to the reduction in signal strength as it travels through a medium, such as a cable or a waveguide. Attenuation can result from factors like distance, obstacles, or the characteristics of the medium itself. To compensate for this loss, amplifiers are used to amplify the signal.

> **Decibel:** The decibel (dB) measures the relative strengths of two signals or one signal at two different points. The decibel is negative if a signal is attenuated and positive if a signal is amplified.

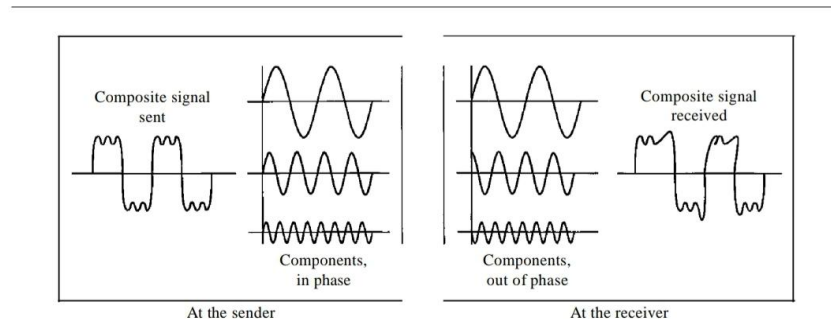
$$\text{dB} = 10 * \log_{10}(\text{power at point2} / \text{power at point1})$$



**Noise:** Noise disrupts the clarity of the transmitted signal and can be caused by various factors, including electrical interference, environmental conditions, or system limitations. Noise can degrade the quality of the transmitted information.



**Distortion:** Distortion alters the shape or quality of the transmitted signal, leading to errors in the received information. This can be caused by factors such as signal reflections, frequency-dependent attenuation, or non-linearities in the transmission medium.



**> Signal to Noise Ratio (SNR):** SNR is the ratio of the signal power to the noise power. A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise.

$$\text{SNR} = \text{Average Signal Power} / \text{Average Noise Power}$$

It is often measured in decibels:

$$\text{SNR}_{\text{db}} = 10 * \log_{10} \text{SNR}$$

**> Types of Noise:**

Thermal Noise: It arises from the random motion of electrons in a conductor. It is present in all electronic circuits and can degrade the quality of the signal.

Crosstalk: This is a type of interference caused by the coupling of signals between communication channels. It can occur in both wired and wireless communication systems, and can lead to errors in data transmission.

Impulse Noise: This type of noise consists of short, unexpected bursts of energy that can disrupt data transmission. It can be caused by sources such as lightning, electrical faults, or external electromagnetic interference.

## Data Rate Limits

Data rate (How fast we send data) depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel another by Shannon for a noisy channel.

**Nyquist Bit Rate:** For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{Bit Rate} = 2 * \text{bandwidth} * \log_2 L$$

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

According to this formula, given a specific bandwidth, we can have any bit rate we want by increasing the number of signal levels. But increasing the levels of a signal may reduce the reliability of the system.

**Shannon Capacity:** In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = \text{bandwidth} * \log_2(1 + \text{SNR})$$

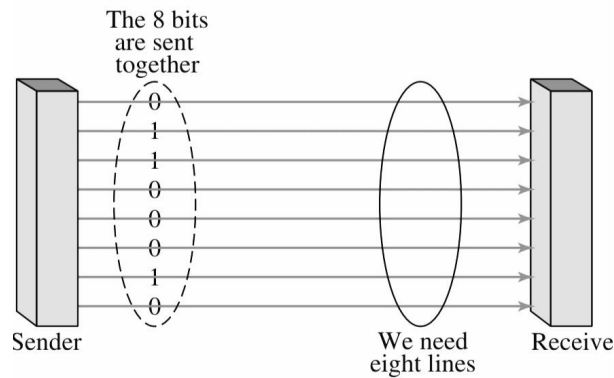
In this formula, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is the capacity of the channel in bits per second. Note that in the Shannon formula there is no indication of the signal level, which means that no matter how many levels we have, we cannot achieve a data rate higher than the capacity of the channel. In other words, the formula defines a characteristic of the channel, not the method of transmission.

In practice, we need to use both methods to find the limits and signal levels. The Shannon capacity gives us the upper limit of the channel; the Nyquist formula tells us how many signal levels we need.

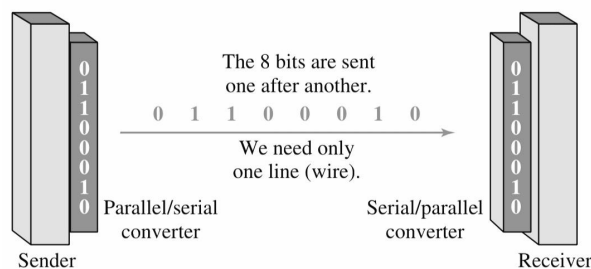
## Transmission Modes

**Parallel Transmission:** Parallel transmission is a method of data transmission where multiple bits are sent simultaneously over separate channels within the same cable or communication link. In parallel transmission, each bit of a byte is transmitted over its own separate wire, allowing for faster data transfer rates compared to serial transmission, where bits are sent sequentially over a single channel. The advantage of parallel transmission is speed. All else being equal, parallel transmission can increase the transfer speed by a factor of n over serial transmission. But there is a significant disadvantage: cost. Parallel transmission requires n communication lines (wires in the example) just to transmit the data stream. Because this is expensive, parallel transmission is usually limited to short distances.





**Serial Transmission:** Serial transmission is a method of data transmission where bits of data are sent sequentially over a single channel. In serial transmission, each bit is transmitted one after the other along the communication channel. The advantage of serial over parallel transmission is that with only one communication channel, serial transmission reduces the cost of transmission over parallel by roughly a factor of  $n$ . Serial transmission occurs in one of three ways: asynchronous, synchronous, and isochronous.



**Asynchronous:** Asynchronous transmission involves data being sent without a shared clock signal, relying on start and stop bits.

**Synchronous:** Synchronous transmission uses a common clock signal to synchronize data transfer.

**Isochronous:** Isochronous transmission ensures data is sent at a constant rate, crucial for real-time applications like audio or video streaming.

## Guided Transmission Media

Guided media, also known as wired or bounded media, refers to the physical channels or pathways through which signals travel in a communication system. Unlike unguided media (such as wireless transmission), guided media provide a physical, tangible medium for the transmission of signals. Types of guided media:

**Twisted Pair Cable:** It Consists of pairs of insulated copper wires twisted together. The twisting helps mitigate electromagnetic interference (EMI) and radio frequency interference

(RFI). The most common UTP connector is RJ45 (RJ stands for registered jack), as shown. The RJ45 is a keyed connector, meaning the connector can be inserted in only one way.

Categories: Comes in various categories (e.g., Cat 5E, Cat 6) with different performance characteristics, including bandwidth and data transmission rates.

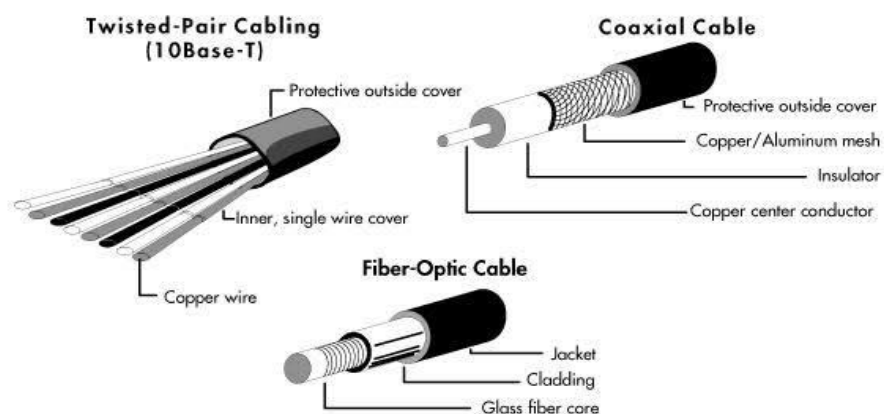
Types: STP cable has a metal foil or braided-mesh covering that encases each pair of insulated conductors. Although metal casing improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive. Unshielded Twisted Pair (UTP) lacks additional shielding.

Applications: Used extensively in telephone systems, local area networks (LANs), and emerging technologies like Power over Ethernet (PoE).

**Coaxial Cable:** It comprises of a central copper conductor, insulating material, metallic shield, and an outer insulating layer. The concentric layers provide enhanced shielding. The most common type of connector used today is the Bayonet Neill-Concelman (BNC) connector.

Types: Various types include RG-59, RG-11 and RG-59.

Advantages: It offers higher bandwidth compared to twisted pair cables, making it suitable for transmitting large amounts of data over longer distances. Better resistance to signal degradation and external interference. It is mostly used in Ethernet LANs.



**Optical Fiber:** It utilizes thin strands of glass or plastic fibers to transmit data as pulses of light. Components include the core (where light travels), cladding (reflective layer), and protective outer layer. Optical fibers use total internal reflection to guide light signal through a channel.

Types: Multimode fibers have a larger core size, allowing multiple light paths (modes) to travel through them. They are often used for shorter distances, such as within a building or a campus. However, they are more susceptible to modal dispersion, where different modes arrive at the destination at different times.

On the other hand, single mode fibers have a smaller core size, supporting only one light path. This makes them ideal for longer-distance transmissions, such as in telecommunications over extensive geographical areas. Single mode fibers are less prone to modal dispersion, enabling higher bandwidth and longer transmission distances compared to multimode fibers.

**Advantages:** Provides exceptional bandwidth, low signal attenuation, and immunity to electromagnetic interference. Light weight, resistance to corrosive materials. Key for long-distance communication, high-speed internet, and emerging technologies like 5G.

## Unguided Transmission Media

Unguided transmission media, also known as unbounded or wireless media, refers to communication channels that do not have a physical conductor or pathway for signals. Instead, these media rely on the propagation of electromagnetic waves through the air or space. Here are some common types of unguided transmission media:

**Radio Waves:** Radio waves are electromagnetic waves (ranging from 3 kHz to 1 GHz) used for wireless communication. Radio waves, for the most part, are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions. Omnidirectional antennas are used to send out Radio signals in all directions.

**Applications:** Used in radio and television broadcasting, wireless data transmission, and mobile communication.

**Microwaves:** Microwaves are higher frequency electromagnetic waves (ranging from 1GHz to 300GHz). They are commonly used for point-to-point communication, such as in microwave ovens and satellite communication. Microwaves need unidirectional antennas that send out signals in one direction. Microwave propagation is line-of-sight. The curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using microwaves. Repeaters are often needed for long distance communication. These cannot penetrate walls.

**Applications:** Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

**Infrared Waves:** Infrared waves have longer wavelengths (ranging from 1 mm to 770 nm) than visible light but shorter than microwaves. Infrared communication is often used for short-range, line-of-sight data transmission.

**Applications:** Infrared remote controls, some short-range wireless communication, Keyboard, PCs. Data rate ranges from 75 kbps to 4 Mbps.

**Satellite Communication:** Involves the use of communication satellites in orbit around the Earth to relay signals between ground-based stations.

**Applications:** Global communication, television broadcasting, and navigation systems like GPS.

# UNIT II

## DATA COMMUNICATION

### Digital-to-Digital Conversion

> **Data Element:** A data element is the smallest entity that can represent a piece of information: this is the bit. In digital data communications, a signal element carries data elements.

> **Signal Element:** A signal element is the shortest unit (timewise) of a digital signal. In other words, data elements are what we need to send; signal elements are what we can send. Data elements are being carried; signal elements are the carriers.

> **Signal Rate:** Also known as baud rate or modulation rate or pulse rate. The signal rate is the number of signal elements sent in 1s. The unit is the baud.

$$\text{Signal Rate} = \text{Bit Rate} / r$$

here  $r$  is the ratio of the number of data elements carried by each signal element

$$S_{\text{avg}} = c * N * (1/r)$$

where  $N$  is the bit rate (bps);  $c$  is the case factor, which varies for each case;  $S$  is the number of signal elements per second.

In analog Transmission,

$$\text{Signal Rate} = \text{Bit Rate} / r$$

where  $r = \log_2 L$  and  $L$  is the number of different signal elements

**Line Coding:** It is a technique used in digital communication to convert a sequence of digital bits into a digital signal. It involves mapping the binary data to a digital signal that can be transmitted over a communication channel, such as a cable or optical fiber.

Different line coding schemes exist, each with its own advantages and disadvantages. Some common examples include Non-Return-to-Zero (NRZ), Manchester encoding, and Bipolar encoding. These schemes help in achieving synchronization, and ensuring reliable data transmission in digital communication systems.

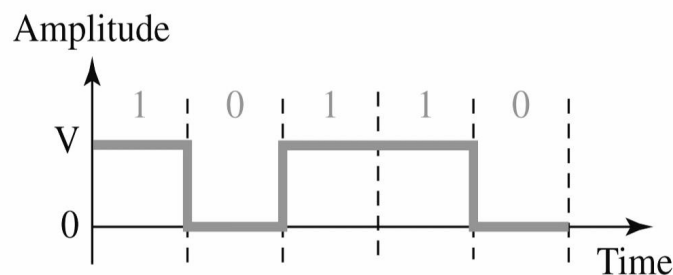
In line coding, polar, unipolar, and bipolar are categories that describe the voltage levels used to represent binary values in the encoded signal.

**1. Unipolar Coding:** Unipolar coding uses only one voltage level to represent binary values. For example In binary 0, the voltage stays at a reference level (e.g., 0V), and for binary 1, the voltage goes to a positive level.

Advantages: Simple implementation, DC component is present.

Disadvantages: May suffer from baseline wander and requires careful consideration of DC components.

i) **NRZ (Non-Return-to-Zero):** a unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0. It is called NRZ because the signal does not return to zero at the middle of the bit. Compared with its polar counterpart, this scheme is very costly.



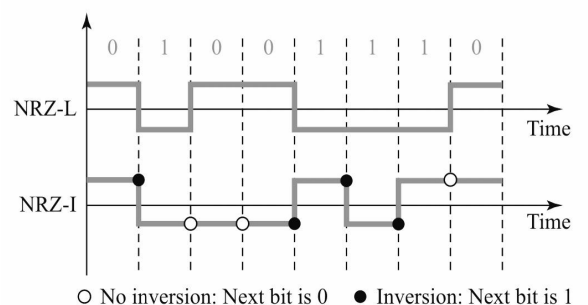
**2. Polar Coding:** Polar coding uses two voltage levels, typically positive and negative, to represent binary values. For example In binary 0, the voltage is at a negative level, and for binary 1, it is at a positive level.

Advantages: No DC component, relatively simple implementation.

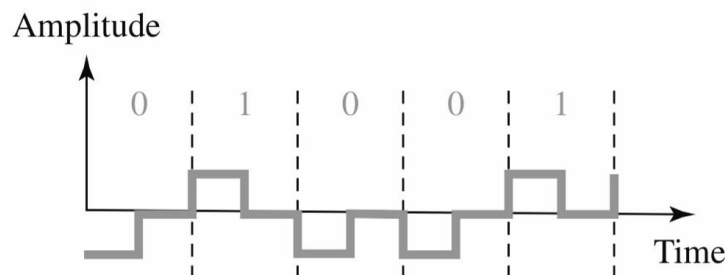
Disadvantages: Potential for baseline wander if proper measures are not taken.

i) **Non-Return-to-Zero Level (NRZ-L):** In NRZ-L the level of the voltage determines the value of the bit. Negative Voltage represents 1 and positive voltage represents 0 or vice-versa. Simple implementation. No need for frequent signal transitions. May have a significant DC component, potentially causing baseline wander.

ii) **Non-Return-to-Zero Invert (NRZ-I):** In Polar NRZ-I, the change or lack of change in the level of the voltage determines the value of the bit. If there is no change, the bit is 0; if there is a change, the bit is 1. Addresses the baseline wander issue of NRZ-L. Provides better synchronization due to more frequent transitions. May still have a DC component, depending on the data.

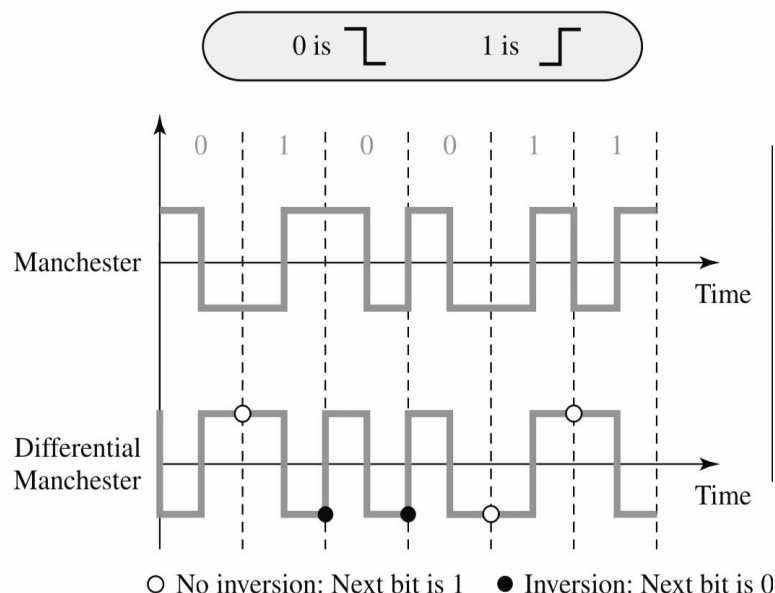


iii) **Return-to-Zero (RZ)**: One solution is the return-to-zero (RZ) scheme, which uses three values: positive, negative, and zero. In RZ, the signal changes not between bits but during the bit. it addresses the problem of non synchronization of sender and receiver clock.



iv) **Manchester**: The idea of RZ and the idea of NRZ-L are combined into the Manchester scheme. In Manchester encoding, the duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half. The transition at the middle of the bit provides synchronization. The Manchester scheme overcomes several problems associated with NRZ-L

v) **Differential Manchester**: It combines the ideas of RZ and NRZ-I. There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit. If the next bit is 0, there is a transition; if the next bit is 1, there is none. The Differential Manchester scheme overcomes several problems associated with NRZ-I.



3. **Bipolar Coding**: Bipolar coding uses three voltage levels: positive, negative, and zero (neutral). For example, Binary 0 is represented by a zero voltage level, and binary 1 alternates between positive and negative levels.

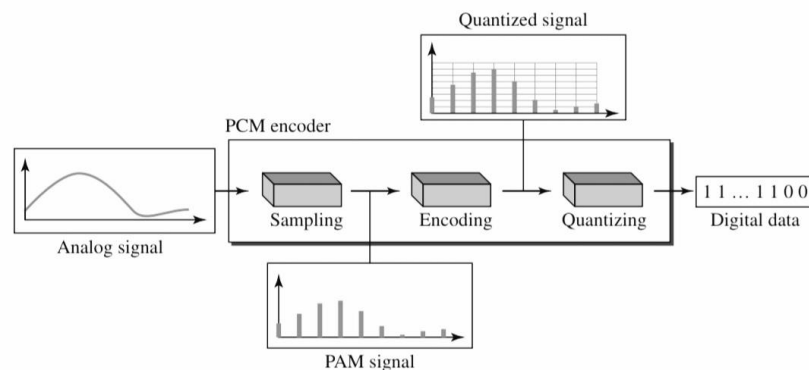
Advantages: No DC component, provides better error detection capability.

Disadvantages: More complex implementation compared to unipolar or polar coding.

## Analog-to-Digital conversion

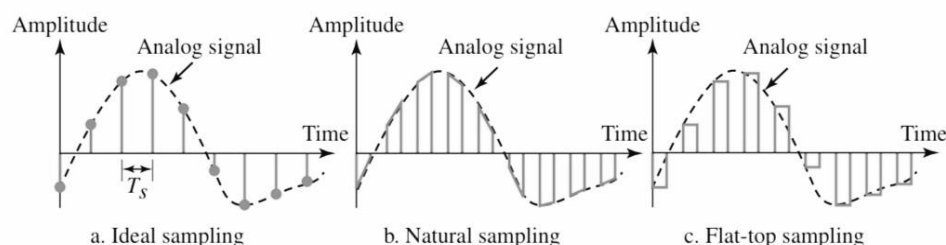
The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM). A PCM encoder has three processes:

1. The analog signal is sampled.
2. The sampled signal is quantized.
3. The quantized values are encoded as streams of bits.



**Sampling:** The first step in PCM is sampling. Sampling is the first step in the PCM process, where the continuous analog signal is sampled at regular intervals. The purpose of sampling is to capture the amplitude of the analog signal at specific points in time. The analog signal is sampled every  $T_s$ s, where  $T_s$  is the sample interval or period. There are three sampling methods—ideal, natural, and flat-top. In ideal sampling, pulses from the analog signal are sampled. This is an ideal sampling method and cannot be easily implemented. In natural sampling, a high-speed switch is turned on for only the small period of time when the sampling occurs. The result is a sequence of samples that retains the shape of the analog signal. The most common sampling method, called sample and hold, however, creates flat-top samples by using a circuit.

The sampling process is sometimes referred to as pulse amplitude modulation (PAM)



**> Sampling Rate:** The inverse of the sampling interval is called the sampling rate or sampling frequency and denoted by  $f_s$ , where  $f_s = 1/T_s$ . According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal.

**Quantization:** Quantization is the process of rounding off the sampled values to the nearest quantization level. The result of sampling is a series of pulses with amplitude values

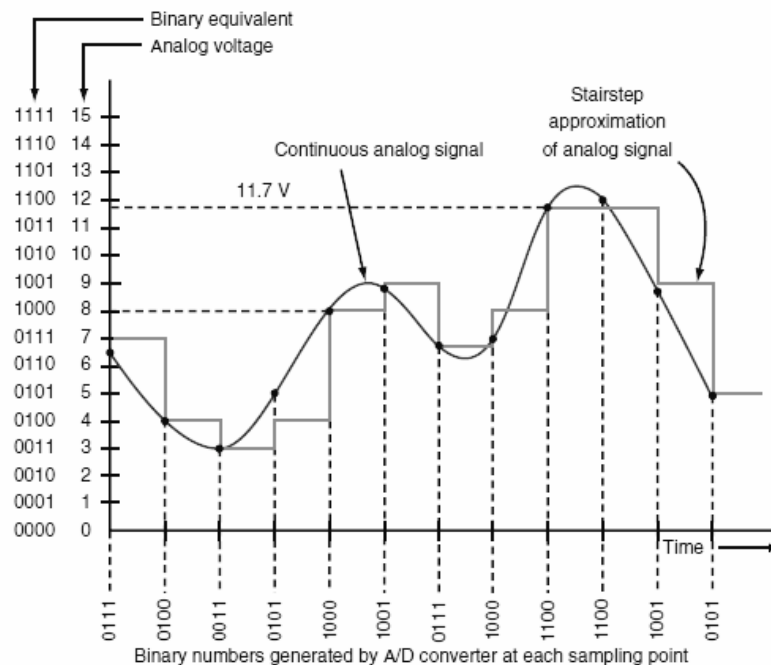
between the maximum and minimum amplitudes of the signal. The set of amplitudes can be infinite with nonintegral values between the two limits. These values cannot be used in the encoding process.

The following are the steps in quantization:

1. We assume that the original analog signal has instantaneous amplitudes between  $V_{\min}$  and  $V_{\max}$ .
2. We divide the range into  $L$  zones, each of height  $\Delta$  (delta).  

$$\Delta = (V_{\max} - V_{\min}) / L$$
3. We assign quantized values of 0 to  $L - 1$  to the midpoint of each zone.
4. We approximate the value of the sample amplitude to the quantized values.

As a simple example, assume that we have a sampled signal and the sample amplitudes are between  $-20$  and  $+20$  V. We decide to have eight levels ( $L = 8$ ). This means that  $\Delta = 5$  V.



**> Quantization Level:** Quantization level refers to the number of discrete levels or steps into which the amplitude of an analog signal is divided during the quantization process. The number of quantization levels directly affects the precision of the digital representation and the fidelity of the reconstructed analog signal. The number of quantization levels determines the bit depth or the number of bits used to represent each sample.

**> Quantization Error:** Quantization error, also known as quantization noise, is the difference between the original analog signal and its quantized digital representation. It occurs because the analog signal is continuous and can have an infinite number of amplitude values, while the quantization process maps these continuous values to a finite set of discrete levels. The quantization error is a type of distortion that arises from the approximation of the analog signal's amplitude to the nearest quantization level during the quantization process.



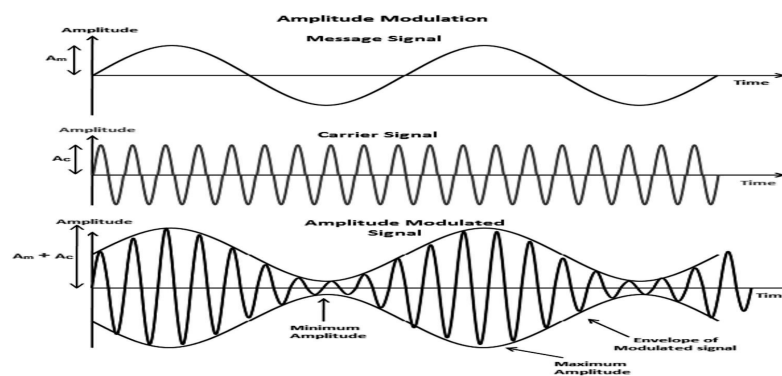
**Encoding:** The last step in PCM is encoding. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an  $n_b$ -bit code word. A quantization code of 2 is encoded as 010; 5 is encoded as 101; and so on. Number of bits for each sample is determined from the number of quantization levels. If the number of quantization levels is  $L$ , the number of bits is  $n_b = \log_2 L$ . If for example  $L$  is 8 and  $n_b$  is therefore 3. The bit rate can be found from the formula:

$$\text{Bit rate} = \text{Sampling rate } (f_s) * \text{number of bits per sample } (n_b)$$

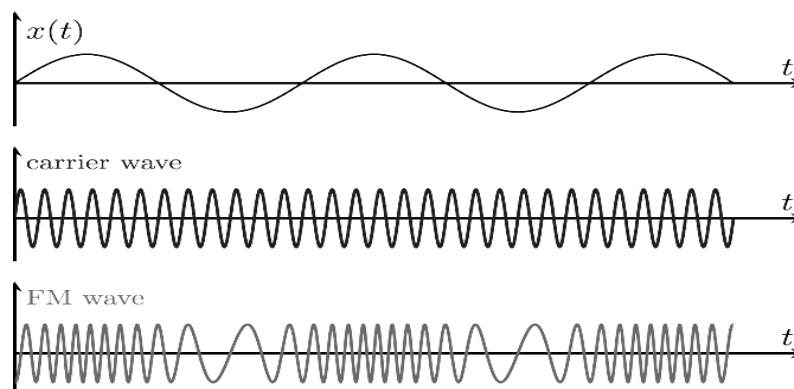
## Modulation and Demodulation

**Modulation:** Modulation is defined as the process of superimposing a low-frequency signal on a high-frequency carrier signal. Modulation is the technique of varying a carrier signal's properties, like amplitude, frequency, or phase, to encode information for transmission. The device used for the process of Modulation is called modulator. There are different modulation types, each altering specific aspects of the carrier signal.

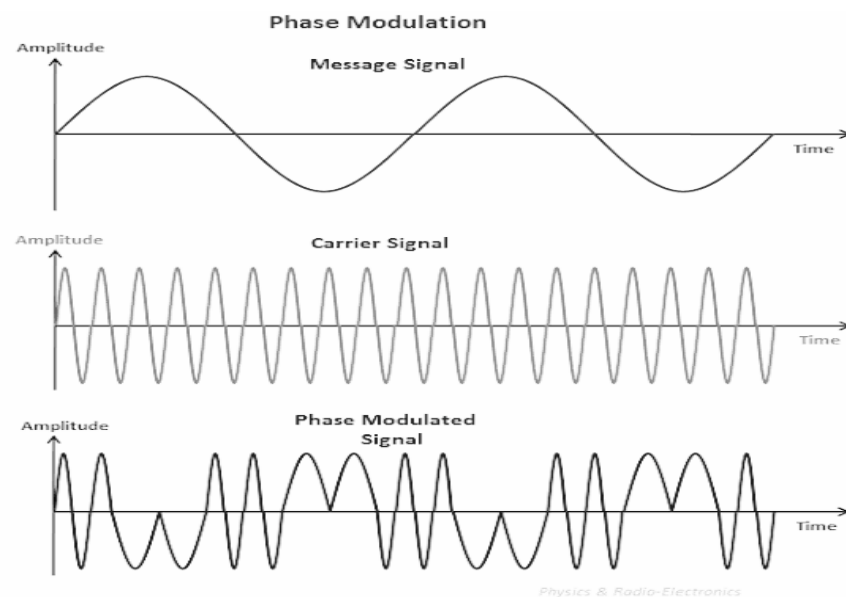
Amplitude Modulation (AM): In Amplitude Modulation varied amplitude represents information. The audio signal changes the carrier signal's amplitude. It is simple and used in radio broadcasting.



Frequency Modulation (FM): In Frequency varied frequency encodes information. Changes in carrier signal frequency reflect the input signal. Common in high-quality audio transmission and radio broadcasting.



**Phase Modulation (PM):** In Phase Modulation varied phase carries information. shifts in carrier signal phase correspond to input signal changes. It is used in certain communication systems and digital data transmission.



#### > Need of Modulation:

1. By using Modulation technique the size of antenna is reduced. Size of antenna is inversely proportional to frequency of signal.
2. Modulation provides a wireless connection to transmit the signals to a longer distance.

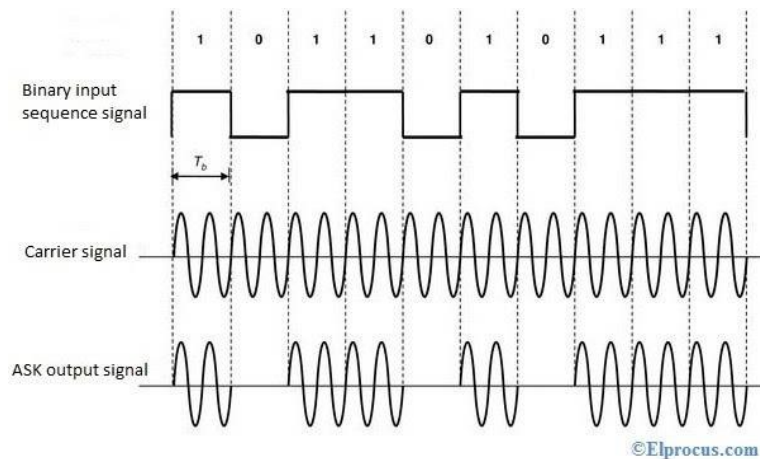
**Demodulation:** Demodulation is defined as extracting the original information-carrying signal from a modulated carrier wave. A demodulator is an electronic circuit that is mainly used to recover the information content from the modulated carrier wave.

Modem (**mo**-Modulator **dem**-Demodulator) is a device that can modulate as well as demodulate a signal.

### Digital-to-Analog conversion

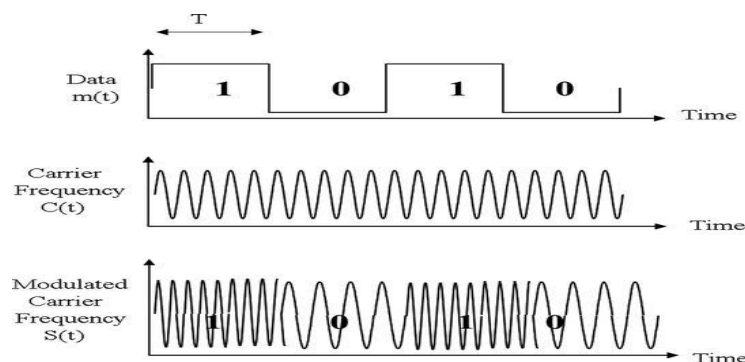
**Amplitude Shift Keying:** In amplitude shift keying, the amplitude of the carrier signal is varied to create signal elements. Both frequency and phase remain constant while the amplitude changes. ASK is normally implemented using only two levels. This is referred to as binary amplitude shift keying or on-off keying (OOK). In multi level Amplitude Shift Keying more than 2 amplitude levels are used.

- In ASK probability of error is high and SNR is less.
- It has lowest noise immunity against noise.
- ASK is a bandwidth efficient system but it has lower power efficiency.



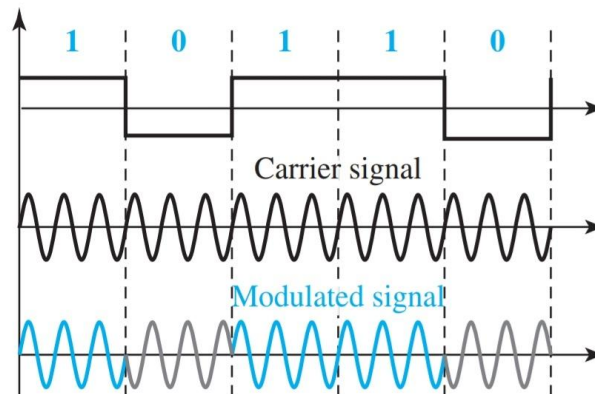
**Frequency Shift Keying:** In frequency shift keying, the frequency of the carrier signal is varied to represent data. The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes. Both peak amplitude and phase remain constant for all signal elements.

- In case of FSK, probability of error is less and SNR is high.
- This technique is widely employed in modem design and development.
- It has increased immunity to noise but requires larger bandwidth compare to other modulation types.



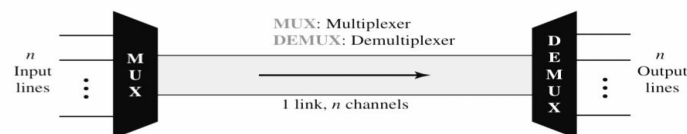
**Phase Shift Keying:** In phase shift keying, the phase of the carrier is varied to represent two or more different signal elements. Both peak amplitude and frequency remain constant as the phase changes. Today, PSK is more common than ASK or FSK. However, QAM (Quadrature Amplitude Modulation), which combines ASK and PSK, is the dominant method of digital-to-analog modulation. Peak Amplitude and Phase remain constant for all signal elements.

- In case of PSK probability of error is less. SNR is high.
- It is a power efficient system but it has lower bandwidth efficiency.
- PSK modulation is widely used in wireless transmission.



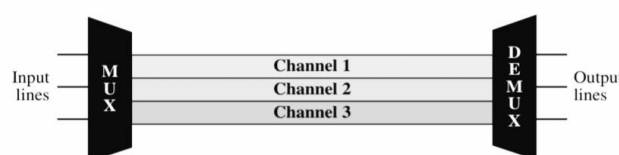
## Multiplexing

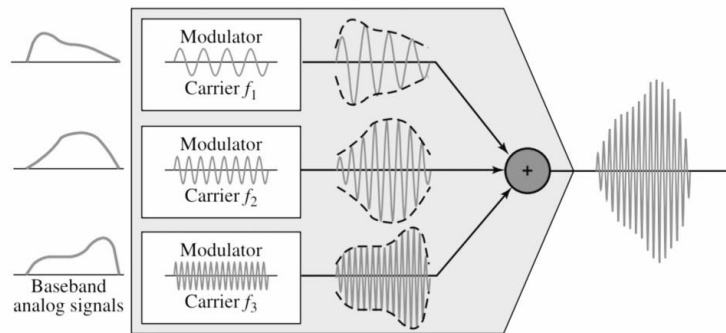
It refers to the process of combining multiple analog or digital signals into one single signal for transmission over a shared medium, like a cable, fiber optic line, or radio wave. This technique allows for efficient use of limited resources, enabling multiple users or devices to share a single communication channel.



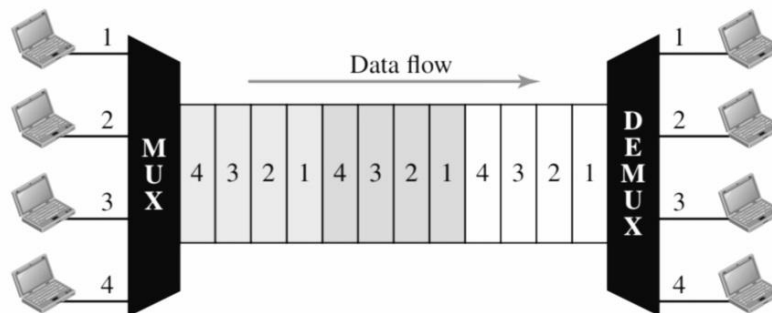
In a multiplexed system,  $n$  lines share the bandwidth of one link. A multiplexer (MUX), combines them into a single stream (many-to-one). At the receiving end, that stream is fed into a demultiplexer (DEMUX), which separates the stream back into its component transmissions (one-to-many) and directs them to their corresponding lines. One link can have many ( $n$ ) channels.

**Frequency Division Multiplexing:** Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted. In FDM, signals generated by each sending device modulate different carrier frequencies. These modulated signals are then combined into a single composite signal that can be transported by the link. Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal. These bandwidth ranges are the channels through which the various signals travel. Channels can be separated by strips of unused bandwidth—guard bands—to prevent signals from overlapping. In addition, carrier frequencies must not interfere with the original data frequencies.



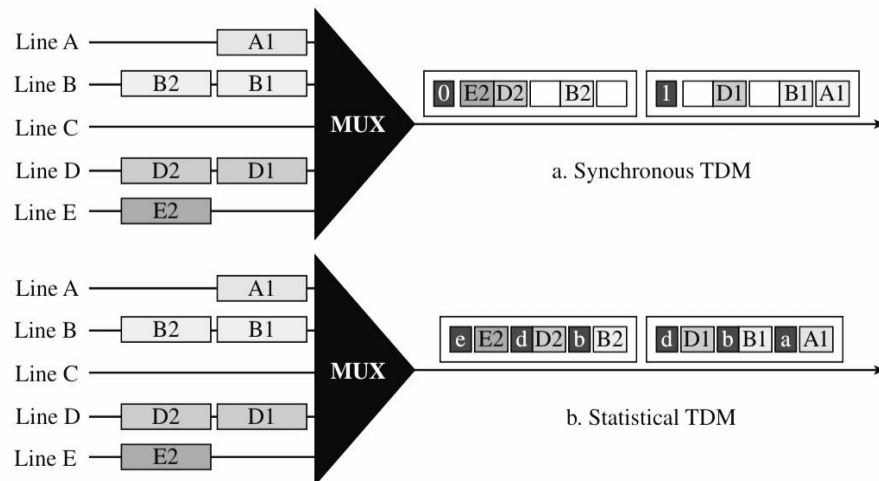


**Time Division Multiplexing:** Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a link. Instead of sharing a portion of the bandwidth as in FDM, time is shared. Each connection occupies a portion of time in the link. Here, the link is sectioned by time rather than by frequency.

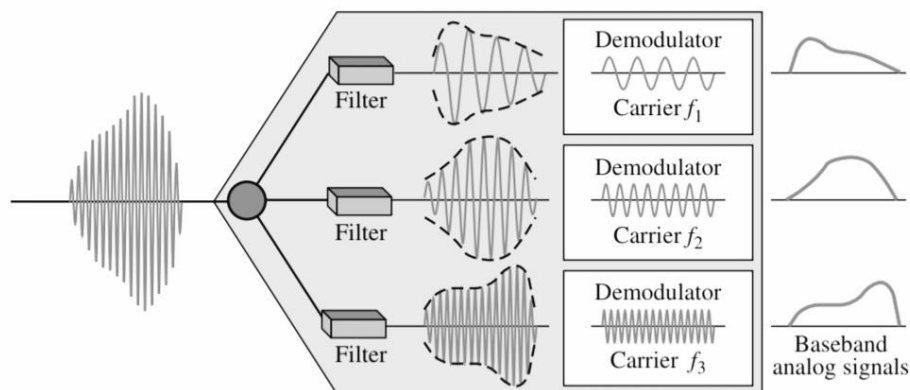


**Synchronous Time Division Multiplexing:** In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot. A unit can be 1 bit, one character, or one block of data. Each input unit becomes one output unit and occupies one output time slot. However, the duration of an output time slot is  $n$  times shorter than the duration of an input time slot. This can be inefficient if some input lines have no data to send.

**Statistical Time Division Multiplexing:** In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency. Only when an input line has a slot's worth of data to send is it given a slot in the output frame. In statistical multiplexing, the number of slots in each frame is less than the number of input lines. The multiplexer checks each input line in round-robin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line.



**De-Multiplexing:** De-multiplexing is the process of separating multiple signals that have been combined into a single channel or transmission path. The demultiplexer uses a series of filters to decompose the multiplexed signal into its constituent component signals. The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the output lines



# UNIT III

## NETWORKING

### Components of Network

A network is composed of various components that work together to enable communication and data exchange between devices. Some of the key components of a network include:

Network Interface Cards (NICs): NICs are hardware components installed in devices that allow them to connect to a network. They provide a physical interface for devices to connect to network cables or wireless networks. MAC (Media Access Control) address is assigned to the network interface card of a device by the manufacturer to uniquely identify a device.

Cables and Connectors: Cables and connectors are used to physically connect devices in a network. Examples include Ethernet cables, fiber optic cables, and connectors such as RJ-45 for Ethernet.

Servers: Servers are computers that store, process, and manage network resources, providing services to other devices.

Hubs: Hubs are basic networking devices that connect multiple devices in a network, but they operate at the physical layer, simply forwarding data to all connected devices.

Switches: Switches are devices that connect multiple devices within a network. They receive network packets and forward them to the appropriate devices based on their target MAC addresses. Switches help create local area networks (LANs) by enabling communication between devices within the same network. A switch operates on the data-link layer of OSI Model.

Routers: Routers are networking devices that connect multiple networks together. They analyze network packets and determine the best path for data to travel between different networks. Routers enable communication between devices in different LANs or wide area networks (WANs). A router operates on the network layer of OSI Model.

Firewalls: Firewalls are security devices that monitor and control incoming and outgoing network traffic. They enforce security policies to protect networks from unauthorized access, malware, and other threats.

Network Protocols: Network protocols are a set of rules and standards that define how devices communicate and exchange data in a network. Examples include TCP/IP (Transmission Control Protocol/Internet Protocol), Ethernet, Wi-Fi (Wireless Fidelity), and DNS (Domain Name System).

## Topologies

Network topologies refer to the physical or logical layout of a network. They determine how devices are interconnected and how data flows within the network. Here are some common network topologies:

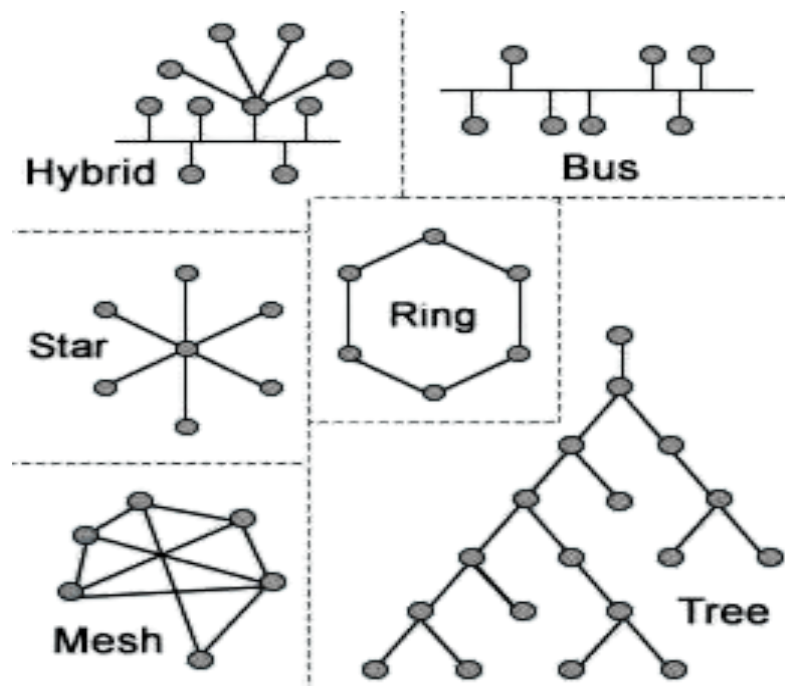
**Star Topology:** In a star topology, all devices are connected to a central device, such as a switch or hub. Each device has a dedicated connection to the central device. This topology provides easy management and troubleshooting, as a single point of failure only affects the connected device.

**Bus Topology:** In a bus topology, all devices are connected to a single cable called a "bus." Each device taps into the bus to send or receive data. However, if the main bus cable fails, the entire network can be disrupted.

**Ring Topology:** In a ring topology, devices are connected in a circular manner, forming a closed loop. Each device is connected to two neighboring devices, and data flows in one direction around the ring. If one device or cable fails, it can disrupt the entire network.

**Mesh Topology:** In a mesh topology, every device is connected to every other device in the network. This creates multiple redundant paths for data to travel, increasing reliability and fault tolerance. Mesh topologies can be full mesh (every device is directly connected to every other device) or partial mesh (only selected devices have multiple connections).

**Tree Topology:** Also known as a hierarchical topology or a hierarchical tree, this topology resembles a tree structure. It consists of multiple interconnected star topologies. Devices are organized in layers, with a central device connecting different branches or levels. This topology is often used in large networks to divide them into manageable segments.





## Categories of Networking

Networking can be broadly categorized into three main types: LAN (Local Area Network), WAN (Wide Area Network), and MAN (Metropolitan Area Network). These categories describe the geographical scope and scale of a network.

**LAN (Local Area Network):** A Local Area Network is a network that covers a small geographical area such as a home, office building, or campus. LANs are typically privately owned and operated by a single organization. They connect devices in close proximity to each other, allowing for high-speed data transfer and resource sharing. LANs often use Ethernet or Wi-Fi technologies. LANs can be wired (using Ethernet cables) or wireless (using Wi-Fi). Examples of LANs include home networks, office networks, and school networks.

Key characteristics of LANs:

1. Limited geographic area
2. High data transfer speeds
3. Privately owned and operated
4. Typically used for local resource sharing and communication

**WAN (Wide Area Network):** A Wide Area Network is a network that spans a large geographical area, connecting multiple LANs or other networks across different locations. WANs are used to connect devices that are far apart, such as different offices in different cities or countries. WANs use various technologies, such as leased lines, fiber optics, satellites, and the internet, to transmit data over long distances. The internet itself is the largest example of a WAN.

Key characteristics of WANs:

1. Wide geographic coverage
2. Lower data transfer speeds compared to LANs
3. Connects multiple LANs or networks
4. Utilizes various technologies for long-distance communication

**MAN (Metropolitan Area Network):** A Metropolitan Area Network is a network that covers a larger geographical area than a LAN but smaller than a WAN. It typically spans a city or a metropolitan area. MANs are designed to provide connectivity for organizations within a specific region. MANs often utilize high-speed fiber optic cables and provide faster data transfer rates compared to WANs. They are commonly used by service providers, educational institutions, and government organizations.

Key characteristics of MANs:

1. Covers a metropolitan area or city
2. Faster data transfer speeds compared to WANs
3. Connects multiple LANs or networks within a specific region
4. Utilizes fiber optic cables for high-speed communication

## Network Models

The networking model describes the architecture, components, and design used to establish communication between the source and destination systems. It provides a common framework for network devices and software to communicate with each other. These models use protocol layering that is, communication protocols are organized into layers, each responsible for specific functions. This layering helps in modularizing the networking tasks and promoting interoperability. The layers interact with adjacent layers, providing a structured approach to network communication.

**OSI Model:** The OSI (Open Systems Interconnection) reference model is a conceptual framework that standardizes the functions of a communication system into seven distinct layers. The model was developed by the International Organization for Standardization (ISO) in 1984 to facilitate interoperability between different vendor devices and protocols. There are the seven layers of the OSI model, from the bottom to the top:

Physical Layer: The physical layer is responsible for transmitting raw bitstream data over physical media, such as copper wires, fiber optics, or wireless signals. It defines the electrical, mechanical, and procedural aspects of physical communication.

Data Link Layer: The data link layer provides error-free transmission of data frames between directly connected devices on a network. It manages data framing, error detection, and flow control. Ethernet switches and wireless access points operate at this layer.

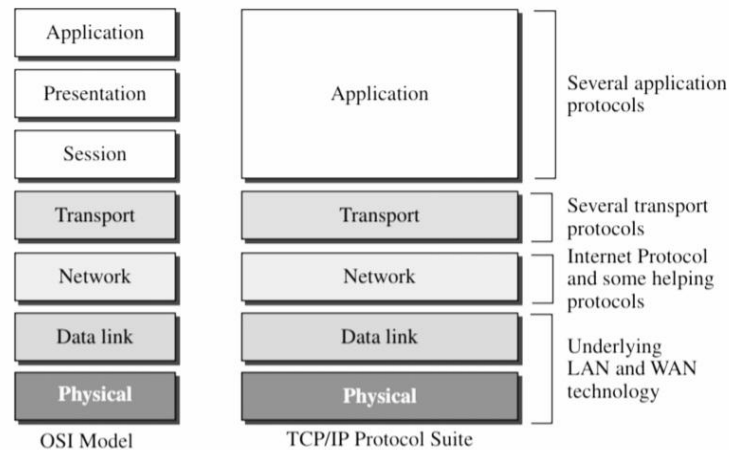
Network Layer: The network layer handles the routing of data packets across multiple networks. It selects the best path for data transmission, applies logical addressing, and performs packet forwarding. Routers operate at this layer.

Transport Layer: The transport layer ensures reliable and error-free end-to-end data delivery between hosts. It segments data from the upper layers into smaller units, manages acknowledgments and retransmissions, and provides flow control. TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) operate at this layer.

Session Layer: The session layer establishes, manages, and terminates communication sessions between applications. It handles synchronization, checkpointing, and recovery of data exchange. It enables session establishment and synchronization between devices.

Presentation Layer: The presentation layer is responsible for data representation, encryption, and compression. It ensures that data is in a format that the application layer can understand. This layer handles tasks like data encryption, decryption, and protocol conversion.

Application Layer: The application layer provides services directly to end-user applications. It enables communication between user applications and network services. Protocols like HTTP (Hypertext Transfer Protocol), FTP (File Transfer Protocol), and SMTP (Simple Mail Transfer Protocol) operate at this layer.



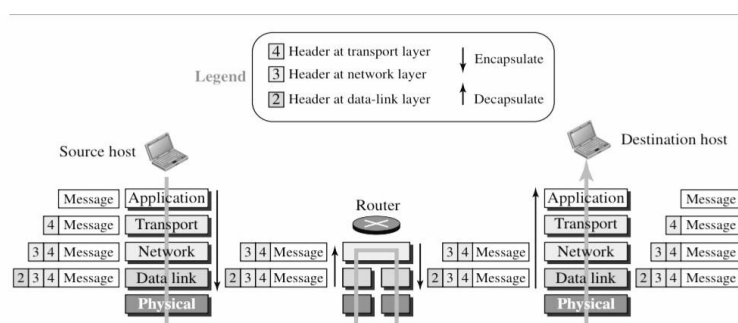
**TCP/IP Protocol Suite:** Introduced by Vint Cerf and Bob Kahn in 1974, TCP/IP model is widely used in the implementation of the internet. It is a more practical and simplified model compared to the OSI model. There are the five layers of the TCP/IP protocol suite.

Physical Layer: The physical layer is responsible for transmitting raw bitstream data over physical media, such as copper wires, fiber optics, or wireless signals. It defines the electrical, mechanical, and procedural aspects of physical communication.

Network Interface Layer (also known as the Link Layer): The Network Interface Layer is responsible for the physical transmission of data packets over the network media. It includes protocols and standards that define how data is formatted, addressed, and transmitted over the physical network, such as Ethernet or Wi-Fi.

Internet Layer: The Internet Layer is where the IP (Internet Protocol) resides. It handles the routing of data packets across networks, addressing, and fragmentation of packets. The IP protocol enables the transmission of data from source to destination over an interconnected network. This layer corresponds to the Network layer in the OSI model.

Transport Layer: The Transport Layer provides end-to-end communication between hosts. It includes protocols like TCP (Transmission Control Protocol) and UDP (User Datagram Protocol). TCP provides reliable, connection-oriented, and error-checked data delivery, while UDP provides best-effort, connectionless data delivery. This layer corresponds to the Transport layer in the OSI model.



Application Layer: The Application Layer is where applications and services directly interact with the network. It includes protocols such as HTTP, FTP, SMTP, DNS, and many others. These protocols enable specific network services and applications to communicate over the network. This layer corresponds to the Session, Presentation, and Application layers in the OSI model.

## Switching

Switching refers to the process of forwarding data packets between devices in a network. It involves the use of network switches, which are devices that connect multiple devices within a network and facilitate the efficient transmission of data.

There are primarily two types of switching:

**Circuit Switching**: Circuit switching establishes a dedicated communication path between two devices for the duration of a communication session. This path remains open throughout the conversation. It was commonly used in traditional telephone networks. However, it is less prevalent in modern data networks.

**Packet Switching**: Packet switching breaks data into smaller packets and routes them individually over the network. Each packet contains a portion of the data, along with addressing information. These packets are then independently transmitted and reassembled at the destination. Packet switching allows for more efficient and shared use of network resources.

**Message Switching**: It involves messages routed in their entirety, one node at a time. Message is stored then routed to next node. Unlike packet switching, the message is not divided into smaller units and sent independently over the network.

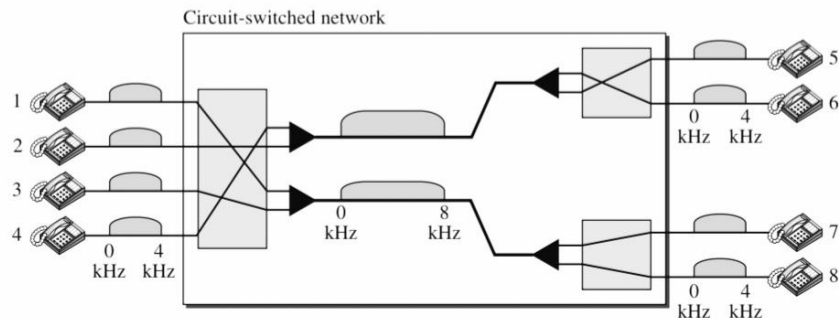
Switches use MAC address tables or forwarding tables to determine where to send incoming packets. By learning the MAC addresses of devices connected to its ports, the switch builds a table that associates MAC addresses with specific ports. This allows the switch to forward incoming packets directly to the destination device without flooding the entire network.

Switching offers several benefits, including:

1. Increased network efficiency: Switches provide dedicated bandwidth to connected devices, allowing for simultaneous communication between multiple devices without collisions.
2. Improved performance: Switches can perform hardware-based forwarding, which enables faster packet processing and reduces latency compared to software-based forwarding.
3. Enhanced security: Switches isolate network traffic by forwarding packets only to the intended recipient, increasing network security and reducing the risk of unauthorized access.
4. Scalability: Switches can be interconnected to create larger networks and accommodate more devices, providing scalability as the network grows.

## Circuit Switched Networks

Circuit-switched networks are a type of telecommunication network that establishes a dedicated communication path between two devices for the duration of a communication session. In circuit switching, a connection is established before any data transmission occurs, and the connection remains open for the entire duration of the communication session.



Key characteristics of circuit-switched networks:

**Dedicated Connection:** When a call is made in a circuit-switched network, a dedicated physical path, also known as a circuit, is established between the calling and receiving devices. This circuit remains allocated exclusively to that communication session until the call is terminated.

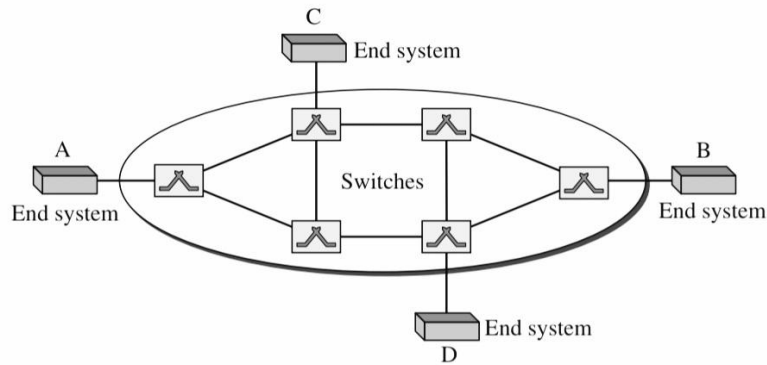
**Constant Bandwidth:** In circuit switching, the allocated bandwidth remains constant, regardless of whether data is being transmitted or not. This ensures a consistent quality of service throughout the call.

**Inefficient Resource Utilization:** Circuit-switched networks allocate resources for the entire duration of the call, even if there are periods of silence or no data transmission.

**Limited Scalability:** Circuit-switched networks are designed for point-to-point communication and are less suitable for connecting multiple devices or locations. Adding more participants to a call would require additional circuits and resources.

## Virtual Circuit Networks

Virtual Circuit Networks (VCNs) are a type of network that combines the benefits of circuit-switched networks and packet-switched networks. VCNs provide a logical or virtual connection between devices, similar to circuit-switching, while utilizing packet-switching techniques for data transmission.



Key features of Virtual Circuit Networks:

Logical Connections: In a VCN, logical connections, known as virtual circuits, are established between devices. These virtual circuits are similar to dedicated physical circuits used in circuit-switched networks but are implemented using packet-switching techniques.

Circuit Identifier: Each virtual circuit is assigned a unique identifier, known as a circuit ID. This circuit ID is used to distinguish between different virtual circuits within the network.

Packet Switching for Data Transmission: Establishment of a virtual circuit is required before data transmission can occur. Once the virtual circuit is established, data transmission occurs using packet-switching techniques. Data is divided into packets, each containing a portion of the data along with addressing information.

Packet Headers: Each packet in a VCN contains a header that includes the circuit ID, source, and destination addresses. This header allows intermediate network devices to identify the correct virtual circuit and route the packets accordingly.

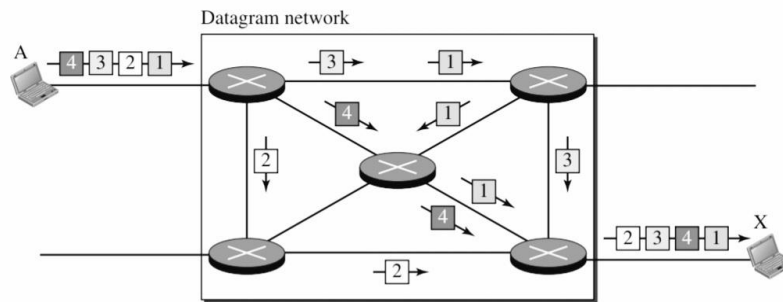
Resource Sharing: Unlike traditional circuit-switched networks, VCNs allow for efficient resource utilization. The network resources, such as bandwidth and routing capacity, can be shared among multiple virtual circuits.

Quality of Service (QoS): VCNs can provide QoS mechanisms to prioritize certain types of traffic within virtual circuits. This allows for differentiated treatment of data and ensures that real-time or critical data receives appropriate network resources.

MPLS Technology: Multi-Protocol Label Switching (MPLS) is commonly used to implement VCNs. MPLS adds labels to packets to create virtual circuits and provides efficient forwarding based on these labels.

## Datagram Networks

Datagram networks are a type of computer network that operates based on the datagram packet switching technique. In a datagram network, data is transmitted in the form of individual packets, called datagrams, which are treated independently and routed separately through the network.



Key features of datagram networks:

Packet-based Transmission: In a datagram network, data is divided into small packets, each containing a portion of the data along with addressing information. These packets are individually transmitted across the network and can follow different paths to reach the destination.

Connectionless Communication: Datagram networks are connectionless, meaning that there is no need to establish a dedicated connection before data transmission. Each packet is treated independently and does not rely on any previous packets or connection setup.

Packet Headers: Each datagram packet in a datagram network contains a header that includes the source and destination addresses, as well as other control information. These headers allow network devices such as routers to route packets based on the destination address.

Best Effort Delivery: Datagram networks typically provide best effort delivery, where packets are delivered to the destination without any guarantees of reliability or ordering. Packets may be lost, duplicated, or delivered out of order, and it is the responsibility of higher-layer protocols to handle these issues if necessary.

Internet Protocol (IP): The most well-known example of a datagram network is the Internet, which is based on the Internet Protocol (IP). IP is a connectionless protocol that operates at the network layer of the TCP/IP protocol suite and is responsible for routing datagram packets across different networks.

Flexibility and Scalability: Datagram networks offer flexibility and scalability since each packet is treated independently. This allows for efficient utilization of network resources and the ability to handle varying traffic patterns and network conditions.

## Routing

Routing is a fundamental concept in computer networking that involves the process of determining the optimal path for data packets to travel from a source to a destination across a network based on the destination address contained in the packet headers. It is a crucial function that enables communication between different devices and networks, allowing data to be transmitted efficiently and accurately.

Routing Protocols: Routing protocols are sets of rules and algorithms used by network devices, such as routers, to exchange routing information and determine the best path for packet forwarding. Examples of popular routing protocols include Border Gateway Protocol (BGP), Open Shortest Path First (OSPF), and Routing Information Protocol (RIP).

Routing Tables: Each router maintains a routing table, which is a data structure containing information about available network paths and their associated destinations. The routing table helps routers determine the next hop to forward packets based on the destination IP address.

Forwarding Decision: When a router receives a packet, it examines the destination IP address in the packet header. Using the routing table, the router determines the next hop (i.e., the next router or network interface) to which the packet should be forwarded. This decision is based on the network topology and the routing protocols in use.

### > **Types of Routing:**

Static routing and Dynamic routing: Static routing involves manually configuring the routing table entries, specifying the next hop for each destination. Dynamic routing, on the other hand, utilizes routing protocols to automatically update and exchange routing information among routers to dynamically adjust the routing decisions.

Interior and Exterior Routing: Interior routing protocols are used within an autonomous system (AS) or a single network, while exterior routing protocols are used to exchange routing information between different networks.



# UNIT IV

## NETWORKING

### Network Addressing

Network addressing is a crucial aspect of computer networking that involves assigning unique identifiers to devices and networks to enable communication and data transfer. It helps in identifying the source and destination of data packets within a network or across different networks. There are two primary types of network addressing: Logical addressing and Physical addressing.

**Physical Addressing:** Physical network addressing refers to the addressing scheme used to uniquely identify devices on a network at the hardware level. This type of addressing is typically associated with the Data Link Layer (Layer 2) of the OSI model. MAC (Media Access Control) address is a unique identifier assigned to the network interface card (NIC) of a device and are assigned by the manufacturer. These are 48-bit addresses expressed as six groups of two hexadecimal digits, separated by colons or hyphens (e.g., 00:1A:2B:3C:4D:5E). Physical addresses are used for local communication within the same network segment.

**Logical Addressing:** It refers to the addressing scheme used to uniquely identify devices on a network at the software level. This type of addressing is typically associated with the Network Layer (Layer 3) of the OSI model. Logical addresses are used to route data across different networks and are independent of the hardware used in the network. It uses IP addresses to route data. IP addresses are 32-bit or 128-bit numeric identifiers, represented in either IPv4 (32-bit) or IPv6 (128-bit) formats. IPv4 addresses are written in the form of four sets of decimal numbers separated by periods (e.g., 192.168.0.1), while IPv6 addresses are written in hexadecimal format with colons separating blocks (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334). Logical addresses are used for global communication across different network segments.

**Address Resolution Protocol (ARP):** ARP is a protocol used to map an IP address to its corresponding MAC address in a local network.

When a device wants to send a data packet to another device on the same network, it uses ARP to determine the MAC address of the destination device. ARP maintains a table called the ARP cache, which stores the IP-to-MAC address mappings of devices within the network.

### Subnetting

Subnetting is the process of dividing a network into two or more subnets. Its primary function is to make the routing of data within a network more efficient and secure. Subnetting also helps make better use of IPv4 addresses.

> **Subnet:** A subnet (short of subnetwork) describes the specific part of a network that shares the network's IP address but has a unique subnet address.

> **Subnet Mask:** A subnet mask is a number that distinguishes the network address and the host address within an IP address. It is used to divide an IP address into network and host portions. It is a 32-bit number that uses a contiguous series of 1s followed by a contiguous series of 0s (in binary). For example, a subnet mask of 255.255.255.0 (11111111.11111111.11111111.00000000) in IPv4 means the first 24 bits are network bits, and the remaining 8 bits are host bits

> **Classless Inter-Domain Routing (CIDR):** CIDR, or Classless Inter-Domain Routing, is a method used to allocate and specify IP addresses and their routing on the Internet. It's a way to allocate and specify IP addresses more flexibly than the traditional system based on classes (Class A, B, C).

Format: CIDR notation combines the IP address and the subnet mask into a single string, separated by a slash ("/"). For example, 192.168.1.0/24 means the IP address is 192.168.1.0, and the subnet mask is 255.255.255.0. The number after the slash indicates the length of the network prefix in bits. In the example 192.168.1.0/24, the prefix length is 24, meaning the first 24 bits represent the network portion.

CIDR notation is widely used in IP addressing, especially in routing tables, to make internet routing more efficient and scalable.

### > **Creating a Subnet**

Creating a subnet using subnetting involves dividing an existing network into smaller subnetworks to efficiently manage and use IP addresses. Here's a general overview of the steps involved in creating a subnet using subnetting:

Determine the network address and subnet mask: Start by identifying the existing network address and the subnet mask. For example, if your network address is 192.168.1.0 and the subnet mask is 255.255.255.0, this represents a Class C network with 24 bits for the network portion.

Decide on the number of subnets and hosts: Determine how many subnets you need and how many hosts each subnet should accommodate. This will help you decide on the subnet mask that will meet your requirements.

Choose a new subnet mask: Based on the number of subnets and hosts per subnet, calculate the number of bits needed for the subnet portion of the IP address. For example, if you need 6 subnets, you will need at least 3 bits for the subnet portion ( $2^3 = 8$  subnets, which is more than 6).

Calculate the new subnet mask: Modify the subnet mask to accommodate the new subnetting requirements.

Assign subnet addresses: Calculate the range of IP addresses for each subnet based on the new subnet mask. This involves determining the subnet address, the range of assignable IP addresses within each subnet, and the broadcast address for each subnet.

Example: Suppose need to create 4 subnets, each accommodating at least 30 hosts.

1. Determine the network address and subnet mask:

Network address: 192.168.1.0

Subnet mask: 255.255.255.0 (or /24 in CIDR notation)

2. Choose a new subnet mask: To accommodate 4 subnets, we need at least 2 bits for the subnet portion ( $2^2=4$  subnets). So, the new subnet mask will have 26 bits for the network portion ( $32 - 2 = 30$  bits for hosts).

3. Calculate the new subnet mask: The new subnet mask will be 255.255.255.192 (or /26 in CIDR notation).

4. Assign subnet addresses:

Subnet 1: 192.168.1.0 - 192.168.1.63

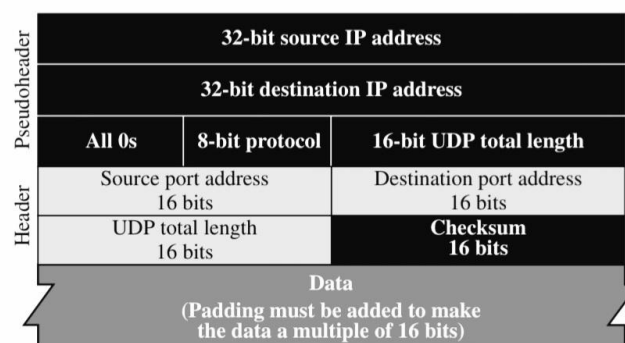
Subnet 2: 192.168.1.64 - 192.168.1.127

Subnet 3: 192.168.1.128 - 192.168.1.191

Subnet 4: 192.168.1.192 - 192.168.1.255 (Subnet addresses, assignable IP ranges, and broadcast addresses are calculated based on the new subnet mask.)

## Transport Layer Protocols

> **User Datagram:** UDP packets, called user datagrams, have a fixed-size header of 8 bytes made of four fields, each of 2 bytes (16 bits). The first two fields define the source and destination port numbers. The third field defines the total length of the user datagram, header plus data. The last field can carry the optional checksum.



**User Datagram Protocol:** UDP is a connectionless protocol, meaning it does not establish a direct connection between the sender and receiver before transmitting data. It provides a simple, lightweight, and low-overhead mechanism for sending data packets over IP networks.

Process-to-Process Communication: UDP provides process-to-process communication using socket addresses, a combination of IP addresses and port numbers.

Connectionless Services: UDP provides a connectionless service. There is no connection establishment and no connection termination. This means that each user datagram sent by UDP is an independent datagram. This means that each user datagram can travel on a different path. Only those processes sending short messages, messages less than 65,507 bytes (65,535 minus 8 bytes for the UDP header and minus 20 bytes for the IP header), can use UDP.

Flow Control: UDP is a very simple protocol. There is no flow control, and hence no window mechanism. The receiver may overflow with incoming messages.

Error Control: There is no error control mechanism in UDP except for the checksum. This means that the sender does not know if a message has been lost or duplicated. When the receiver detects an error through the checksum, the user datagram is silently discarded.

> **Segment (TCP)**: In TCP, data is transmitted in segments. Each segment contains a header with control information and a payload of data from the upper layers. The TCP header includes information such as source and destination port numbers, sequence number, acknowledgment number, window size, and control flags. The payload of the segment contains the actual data being transmitted. TCP segments are encapsulated within IP packets for transmission over IP networks.

**Transmission Control Protocol**: TCP is a connection-oriented, reliable protocol. TCP explicitly defines connection establishment, data transfer, and connection teardown phases to provide a connection-oriented service. To provide reliability TCP uses checksum (for error detection), retransmission of lost or corrupted packets, cumulative and selective acknowledgments, and timers.

Process-to-Process Communication: TCP provides process-to-process communication using port numbers.

Connection Establishment: Three-way handshake is used to establish a connection: SYN, SYN-ACK, ACK. Ensures both sender and receiver are ready to exchange data.

Reliability: Guaranteed delivery of data through acknowledgment mechanism. Retransmission of lost or corrupted packets.

Flow Control: Prevents overwhelming the receiver with too much data. Uses windowing mechanism to control the flow of data.

Error Detection: Checksum is used to detect errors in transmitted data. If errors are detected, TCP requests retransmission.

Ordered Data Transfer: Ensures data arrives in the correct order at the destination.

Stateful Protocol: Maintains the state of the connection throughout its duration.

### > Key differences between UDP and TCP:

Connection: UDP is connectionless, while TCP is connection-oriented.

Reliability: UDP does not guarantee reliable data delivery, while TCP ensures reliable data delivery and packet sequencing.

Order: UDP does not ensure the order of packet delivery, while TCP guarantees the order of packet delivery.

Overhead: UDP has lower overhead compared to TCP, making it faster and more efficient.

Applications: UDP is commonly used in real-time applications, while TCP is used for applications that require reliable and accurate data delivery.

## IPv4

**IPv4**: IPv4 (Internet Protocol version 4) is the fourth version of the Internet Protocol, which is the underlying protocol used for communication in computer networks. It is the most widely deployed version of IP and has been in use since its inception in the early days of the internet.

Addressing Scheme: IPv4 addresses are 32-bit numbers, represented as four sets of decimal numbers ranging from 0 to 255, separated by periods (e.g., 192.168.0.1).

This allows for a total of approximately 4.3 billion unique IPv4 addresses.

IPv4 addresses are divided into network prefixes and host identifiers, with the network prefix indicating the network portion of the address and the host identifier specifying the specific device within that network.

**Classful Addressing**: Classful addressing refers to the original system of allocating IP addresses based on predefined classes: Class A, Class B, and Class C. These classes had fixed boundaries, and each class had a predetermined range of IP addresses for network and host portions. The classes are as follows:

1. Class A (1.0.0.0 to 126.255.255.255): First octet is the network portion, and the remaining three octets are for hosts. Supports up to 128 networks and 16 million hosts per network. Typically assigned to large organizations.
2. Class B (128.0.0.0 to 191.255.255.255): First two octets represent the network, and the last two octets are for hosts. Supports up to 16,384 networks and 65,536 hosts per network. Suitable for medium-sized organizations.

3. Class C (192.0.0.0 to 223.255.255.255): First three octets are the network, and the last octet is for hosts. Supports up to 2 million networks and 256 hosts per network. Commonly used for small-sized networks.

4. Class D (224.0.0.0 to 239.255.255.255): Reserved for multicast.

5. Class E (240.0.0.0 to 255.255.255.255): Reserved for experimental use.

With the introduction of Classless Inter-Domain Routing (CIDR), which allows for variable-length subnet masking, the strict class boundaries have become more flexible.

> **Network Address Translation (NAT):** Due to the limited availability of IPv4 addresses, Network Address Translation (NAT) is commonly used to allow multiple devices within a private network to share a single public IPv4 address.

NAT translates private IP addresses into a public IP address, enabling devices on the private network to communicate with devices on the internet.

## Application Layer Protocols

**Hypertext Transfer Protocol:** HTTP is an application layer protocol used for transmitting hypertext documents on the World Wide Web.

It operates on the client-server model, where a client (such as a web browser) sends an HTTP request to a server, and the server responds with an HTTP response containing the requested content. It uses TCP as its transport protocol and typically runs on port 80 for unencrypted communication and port 443 for encrypted communication (HTTPS).

HTTP requests are made using methods such as GET, POST, PUT, DELETE, etc., to perform actions like retrieving web pages, submitting form data, updating resources, or deleting resources. It is a stateless protocol, meaning each request-response cycle is independent and does not retain any information about previous interactions. HTTP supports various media types, such as HTML, CSS, JavaScript, images, videos, etc., allowing the retrieval and display of web content.

**File Transfer Protocol:** FTP is a standard network protocol used for transferring files between a client and a server on a computer network.

It operates on the client-server model, where the client establishes a connection with the server and performs file transfer operations. It uses TCP as its transport protocol and typically runs on port 21 for control connection and port 20 for data connection. FTP allows for various file operations, including uploading (put), downloading (get), renaming, deleting, creating directories, etc.

It supports two modes of operation: active mode and passive mode. In active mode, the server establishes a connection back to the client for data transfer, whereas in passive mode, the client establishes a data connection to the server.

FTP supports authentication mechanisms for user login and authorization, including username and password or anonymous login. It is commonly used for website maintenance, uploading files to a server, downloading files from a server, and sharing files between computers on a network.

**Simple Mail Transfer Protocol:** SMTP is an application layer protocol used for sending and receiving email messages between mail servers.

It is responsible for the transmission of email messages over the internet or within an internal network. It operates on the client-server model, where an email client (such as Outlook or Gmail) sends an email to an SMTP server, which then relays the email to the recipient's SMTP server. It uses TCP as its transport protocol and typically runs on port 25.

SMTP provides a set of commands (e.g., HELO, MAIL FROM, RCPT TO, DATA) that are exchanged between the client and server to establish a connection, authenticate, and transmit the email message.

It supports email delivery, routing, and handling of various types of email content, including attachments. SMTP is a store-and-forward protocol, meaning the server can hold and forward emails if the recipient's server is temporarily unavailable.

**Simple Network Management Protocol:** SNMP is an application layer protocol used for managing and monitoring network devices and systems.

It allows network administrators to monitor the performance, availability, and configuration of network devices, such as routers, switches, servers, printers, etc. It operates on the client-server model, where a management station (SNMP manager) collects information from SNMP-enabled devices (SNMP agents) on the network.

It uses UDP as its transport protocol and typically runs on port 161. It employs a hierarchical structure of management information, organized into a Management Information Base (MIB), which defines the data structure and attributes of managed objects.

SNMP allows for the retrieval (GET), modification (SET), and notification (TRAP) of data from SNMP agents. It provides a standardized framework for network monitoring and management, enabling tasks such as device configuration, performance monitoring, and fault detection.

**Domain Name System:** DNS is a fundamental protocol used in computer networks to translate human-readable domain names into IP addresses.

DNS acts as a distributed and hierarchical database system that allows mapping of domain names (such as example.com) to their corresponding IP addresses (such as 192.0.2.1). It eliminates the need for users to remember and use IP addresses directly, making it easier to navigate and access resources on the internet.



DNS operates on a client-server model, where DNS clients (such as web browsers or applications) send DNS queries, and DNS servers respond with the requested information.

When a user enters a domain name in a web browser, the browser sends a DNS query to a DNS resolver (typically provided by the Internet Service Provider or a third-party DNS resolver). The resolver then recursively traverses the DNS hierarchy to find the authoritative DNS server responsible for the domain in question. The authoritative DNS server contains the specific DNS records for the domain and returns the corresponding IP address or other information requested by the resolver.

DNS supports various record types, including A records (for IPv4 addresses), AAAA records (for IPv6 addresses), CNAME records (for aliasing one domain to another), MX records (for email routing), and more.

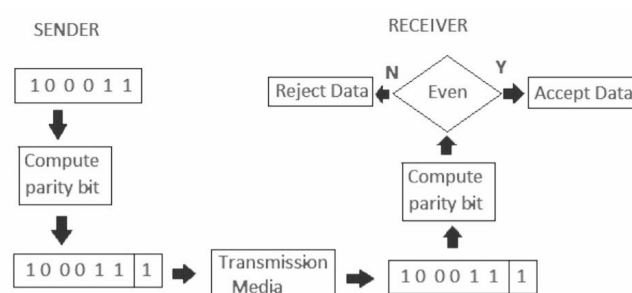
## Error Detection & Correction

Error detection techniques aim to identify if errors have occurred during the transmission of data. Error is a condition when the receiver's information does not match the sender's information. During transmission, digital signals suffer from noise that can introduce errors in the binary bits traveling from sender to receiver. That means a 0 bit may change to 1 or a 1 bit may change to 0. Error correction techniques go beyond error detection and aim to correct errors that are detected during transmission.

**Parity Check:** Parity check is an error detection technique commonly used in digital communication systems. It involves the addition of an extra bit, called the parity bit, to the transmitted data. The purpose of the parity bit is to provide a simple mechanism for detecting errors during data transmission.

Parity can be either even or odd. The sender and receiver must agree on which parity type to use. The sender calculates the parity bit based on the data being transmitted. The most common method is to count the number of bits set to 1 in the data. If the count is even for even parity, the parity bit is set to 0. If the count is odd for odd parity, the parity bit is set to 1.

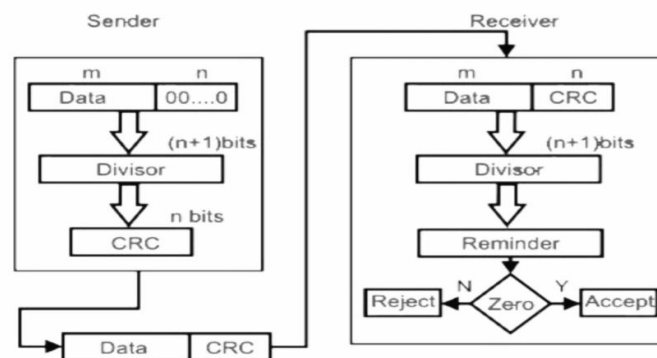
The sender appends the calculated parity bit to the transmitted data. The receiver receives the data along with the parity bit. The receiver performs the same parity calculation on the received data, counting the number of bits set to 1. The receiver compares the calculated parity bit with the received parity bit. If they match, no error is detected. If they differ, an error is detected.





**Disadvantage:** Parity check is not able to detect even no. of bit errors. For example, the Data to be transmitted is 101010. Codeword transmitted to the receiver is 1010101 (we have used even parity). Let's assume that during transmission, two of the bits of code word flipped to 1111101. On receiving the code word, the receiver finds the no. of ones to be even and hence no error, which is a wrong assumption.

**Cyclic Redundancy Check:** CRC is based on binary division. In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of the data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number. At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted. A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.



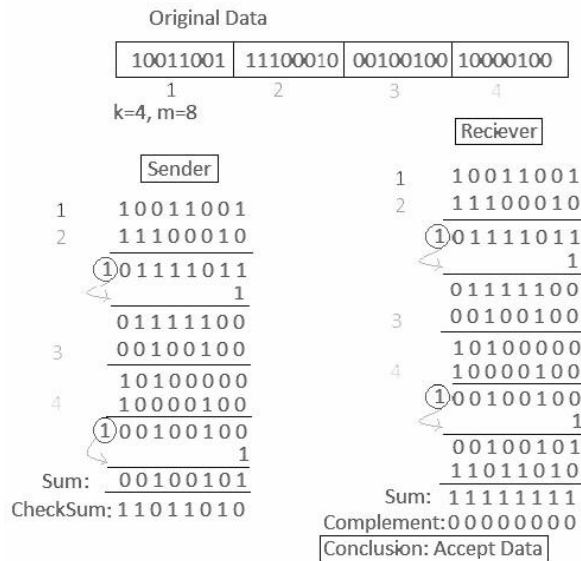
**Disadvantages:** It can identify that errors exist but cannot pinpoint the exact location or correct the errors within the data.

**Checksum:** Checksum error detection is a method used to identify errors in transmitted data. The process involves dividing the data into equally sized segments and using a 1's complement to calculate the sum of these segments. The calculated sum is then sent along with the data to the receiver. At the receiver's end, the same process is repeated and if all zeroes are obtained in the sum, it means that the data is correct.

**Operation at Sender's Side:** Firstly, the data is divided into  $k$  segments each of  $m$  bits. On the sender's end, the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum. The checksum segment is sent along with the data segments.

**Operation at Receiver's Side:** At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented. If the result is zero, the received data is accepted; otherwise discarded.

**Disadvantages:** If one or more bits of a segment are damaged and the corresponding bit or bits of opposite value in a second segment are also damaged.



**Hamming Distance:** Hamming distance is a concept used in information theory and computer science to measure the difference between two strings of equal length. The Hamming distance between two strings is defined as the number of positions at which the corresponding elements in the strings are different.

In order to calculate the Hamming distance between two strings, a and b, we perform their XOR operation,  $(a \oplus b)$ , and then count the total number of 1s in the resultant string.

For example: Suppose there are two strings 1101 1001 and 1001 1101.

$11011001 \oplus 10011101 = 01000100$ . Since, this contains two 1s, the Hamming distance is 2.

**Hamming Code:** Hamming Code is an error-detecting and error-correcting code widely used in digital communication. It adds redundant parity bits to data to detect and correct single-bit errors in transmitted information. By calculating syndromes and performing parity checks, Hamming Code enhances data integrity. It is commonly employed in ECC memory, communication protocols, and data storage, providing a crucial layer of reliability by addressing errors in a systematic manner.

The number of parity bits added to the original data depends on the size of the data block. For example, for a data block of  $n$  bits,  $r$  parity bits are added, where  $r$  is the smallest integer that satisfies the equation  $2^r \geq n + r + 1$ .

Hamming(7,4): Also known as the (7,4) Hamming code, this is one of the simplest and most well-known types of Hamming codes. It operates on 4-bit data blocks and adds 3 parity bits to create 7-bit codewords. This code is capable of detecting 2-bit and correcting single-bit errors.